

Evidence base for the statistical layer

Companion to MEASUREMENT-SPEC.md, CALCULATIONS.md and CHART-SCALING.md. Those documents cite the motor-learning and data-visualisation literature. This one supplies the missing citations for the **statistical machinery** — sample size, quantile estimation, order statistics, and pooling — which until now was stated without sources. Compiled 2026-09-02.

0. Why this document exists

The three existing specs are well sourced on motor learning (Soderstrom & Bjork, Shea & Morgan, Adams, Newell) and on rendering (Tufte, Cleveland). The statistical formulas were not sourced at all. Four of them turn out to have caveats in the primary literature that materially affect this application, which is why the companion document SPEC-REVISIONS.pdf exists.

One body of literature was missing entirely: **sports-science measurement theory**. That field has spent thirty years on the exact problem this app has — deciding whether one athlete's performance actually changed, given noisy repeated tests. It is a closer match than anything else cited so far.

1. Reliability and the smallest worthwhile change

Hopkins (2000), *Sports Medicine* 30(1), 1–15. The standard reference for measurement reliability in athlete testing.

Hopkins establishes that **typical error, expressed as a coefficient of variation, is the appropriate reliability measure** for performance tests, and that systematic between-trial changes reflecting learning, motivation, or fatigue must be removed before estimating within-subject variation. That second clause is the direct justification for the warm-up and re-familiarisation exclusions in CALCULATIONS.md section 4b — they are not an optional refinement, they are a precondition for the CV to mean anything.

Two numbers from this literature matter to the app:

- **Smallest worthwhile change (SWC)** is conventionally 0.2 x between-subject SD, or 0.3 x within-subject SD. For single-athlete tracking the within-subject form applies, giving $SWC = 0.3 \times CV$.
- **Reference CV for athletic performance tests is 1–5%.** Common interpretation thresholds are under 5% good, 5–10% moderate, over 10% poor.

The decision rule is that a test can detect meaningful change only when typical error is smaller than the smallest worthwhile change. This gives a principled value for TARGET_EFFECT, which is currently set to 5% by assertion. See SPEC-REVISIONS section 5.

Note that scenario CVs reported in CALCULATIONS.md (6.8% to 13%) sit at the moderate-to-poor end of that scale. Aim-trainer scores are noisier than most laboratory athletic tests. This is a property of the task, not a flaw in the app, but it sets a hard ceiling on what any amount of clever statistics can extract.

2. Sample size and statistical power

Cohen (1988), *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. The standard two-sample formula for comparing means:

$$\begin{aligned} n_{\text{per_group}} &= 2 * [(z(1-\alpha/2) + z(1-\beta)) / d]^2 \\ &= 2 * (1.96 + 0.84)^2 / d^2 \quad \text{at 80\% power, } \alpha .05 \\ &= 15.68 / d^2 \end{aligned}$$

with $d = \text{delta} / \text{sigma}$, this is $15.7 * (\text{sigma}/\text{delta})^2$

The constant used in CALCULATIONS.md section 4 is correct. The stated alternatives (21.0 for 90% power, 11.3 for 70%) also check out: $2 \times (1.96 + 1.282)^2 = 21.0$ and $2 \times (1.96 + 0.524)^2 = 12.3$, the latter being slightly conservative as written.

The formula assumes independent observations. Runs within one session are not fully independent — they share warm-up state, fatigue, and mood — so the effective sample size is below the raw run count and the required n is an *underestimate*.

3. Order statistics and the volume artifact

Blom (1958), *Statistical Estimates and Transformed Beta Variables*. Royston (1982), Algorithm AS 177. Harter (1961).

MEASUREMENT-SPEC section 4.1 uses Blom's approximation to compute the expected maximum of n normal samples:

$$E[X(r,n)] \sim \mu + \sigma * \text{PhiInv}((r - 0.375) / (n + 0.25))$$

The formula is correctly transcribed. However, comparative evaluations find Blom's approximation performs poorly specifically for the *maximum* in small samples, and is reliable only for reasonably large n , on the order of $n > 100$. Sessions in this app are 15–40 runs. Recommended alternatives, in order of accuracy: exact order statistics computed by numerical integration and cached in a lookup table; a polynomial regression approximation; or Monte Carlo. Royston's AS 177 is the standard exact algorithm and is validated to $n = 2000$.

4. Quantile estimation and its standard error

Harrell & Davis (1982), *Biometrika* 69, 635–640. Maritz & Jarrett (1978). Wilcox, *Introduction to Robust Estimation*.

This is the most consequential section for the current build. The traditional quantile estimator — the one based on one or two order statistics, which is what a plain p_{90} or p_{10} computation uses — is known to have a **relatively high standard error, and the problem is worst when sampling from a light-tailed or normal distribution**. That is exactly this case.

The Harrell-Davis estimator addresses this by using a weighted sum of *all* order statistics, with beta-distribution weights concentrated near the target quantile. It is more efficient than traditional estimators in small samples and asymptotically equivalent to them. Jackknife standard errors are available and are implemented in standard packages.

The trade-off is robustness: because every order statistic carries positive weight, a single corrupted value can influence the estimate. For heavy-tailed data a trimmed variant is preferable. Aim-trainer scores are roughly light-tailed above the warm-up exclusion, so plain Harrell-Davis is appropriate here.

For standard errors, the Maritz-Jarrett method is the standard alternative to density-based formulas. Its documented limitation matters: it needs a minimum count per quantile and **tends to overestimate error below roughly 30 observations** — still preferable to a density estimate built from tail order-statistic spacings, which has no error characterisation at all at these sample sizes.

5. Combining estimates across cells

Hedges & Olkin (1985), *Statistical Methods for Meta-Analysis*. Sanchez-Meca & Marin-Martinez (1998, 2010).

The inverse-variance weighting in CALCULATIONS.md section 3 is correct and is the standard result: **the optimal weight for averaging independent effect sizes is the inverse variance of each estimate**, yielding the minimum-variance unbiased estimator. The pooled standard error follows as the reciprocal square root of the summed weights.

Simulation work comparing inverse-variance weighting against sample-size weighting finds the inverse-variance approach achieves lower mean squared error, at the cost of slight bias. One caveat worth knowing: when the true weights barely differ, simple averaging is preferred for parsimony with negligible efficiency loss. For this app the weights differ substantially between an 8-run cell and an 800-run cell, so the added complexity earns its place.

6. Communicating the result

McGraw & Wong (1992), *Psychological Bulletin* 111, 361–365. Wolfe & Hogg (1971). Grissom (1994).

The probability of superiority proposed in MEASUREMENT-SPEC section 7 is the common-language effect size: the probability that an observation drawn from one group exceeds one drawn from the other. It was introduced by Wolfe and Hogg, revisited by McGraw and Wong under the common-language name, and given the probability-of-superiority label by Grissom. The motivating argument in the original paper is precisely the one the spec makes — that it is far easier for a non-statistical reader to interpret than a standardised difference.

7. Full reference list

Statistical layer — new in this document

1. Hopkins, W. G. (2000). Measures of reliability in sports medicine and science. *Sports Medicine*, 30(1), 1–15. sportsci.org/resource/stats/Hopkins_SportsMed_rely_00.pdf · pubmed.ncbi.nlm.nih.gov/10907753/
2. Hopkins, W. G. A new view of statistics: measures of reliability. sportsci.org/resource/stats/precision.html
3. Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Lawrence Erlbaum.
4. Cohen, J. (1992). A power primer. *Psychological Bulletin*, 112(1), 155–159.
5. Blom, G. (1958). *Statistical Estimates and Transformed Beta Variables*. Wiley.
6. Royston, J. P. (1982). Algorithm AS 177: Expected normal order statistics (exact and approximate). *Applied Statistics*, 31(2).
7. Harter, H. L. (1961). Expected values of normal order statistics. *Biometrika*, 48, 151–165.
8. Branke, J. et al. Order-statistic treatment and Blom accuracy comparison. gwern.net/order-statistic — practical accuracy comparison of twelve methods for the expected maximum.
9. Harrell, F. E., & Davis, C. E. (1982). A new distribution-free quantile estimator. *Biometrika*, 69, 635–640.
10. Maritz, J. S., & Jarrett, R. G. (1978). A note on estimating the variance of the sample median. *JASA*, 73, 194–196.
11. Wilcox, R. R. *Introduction to Robust Estimation and Hypothesis Testing*. Academic Press.
12. Akinshin, A. (2021). Trimmed Harrell-Davis quantile estimator. arXiv:2111.11776. arxiv.org/pdf/2111.11776
13. Hedges, L. V., & Olkin, I. (1985). *Statistical Methods for Meta-Analysis*. Academic Press.
14. Sanchez-Meca, J., & Marin-Martinez, F. (1998). Weighting by inverse variance or by sample size in meta-analysis. *Educational and Psychological Measurement*, 58(2).
15. Marin-Martinez, F., & Sanchez-Meca, J. (2010). Weighting by inverse variance or by sample size in random-effects meta-analysis. *Educational and Psychological Measurement*, 70(1).
16. McGraw, K. O., & Wong, S. P. (1992). A common language effect size statistic. *Psychological Bulletin*, 111, 361–365.
17. Grissom, R. J. (1994). Probability of the superior outcome of one treatment over another. *Journal of Applied Psychology*, 79, 314–316.
18. Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate. *JRSS-B*, 57(1), 289–300. — for the untreated multiple-comparison gap, MEASUREMENT-SPEC gap 9.

Motor learning and rendering — already cited in MEASUREMENT-SPEC section 12

Soderstrom & Bjork (2015); Shea & Morgan (1979); Magill & Hall (1990); Pauwels et al. (2014); Tsay et al. (2023); Wulf & Shea (2002); Adams (1952); Newell et al. (2013); Tufte (1983); Cleveland (1993). Not repeated here.

8. What is still unsourced

Stated plainly, so it is not mistaken for evidence:

- **The 60-minute session gap.** SESSION_GAP_MIN is a reasonable convention with no citation behind it. It should be validated against the user's own inter-run interval distribution, which is bimodal and will show the natural break.
- **REFAM_GAP_DAYS = 14 and REFAM_DROP = 5.** Warm-up decrement is well established (Adams 1952) but its time constant for this task is unmeasured. These are placeholders.
- **CM_CLUSTER_RATIO = 1.1.** No literature on the perceptual threshold for a meaningful sensitivity difference exists. The 10% figure is a design choice.

- **The Recommended-to-play scoring weights.** Entirely heuristic, and should not be presented as anything else.

None of these are errors. They are conventions, and the honest position is that they are conventions awaiting per-install calibration — which is what MEASUREMENT-SPEC section 11 already argues for on different grounds.